Computer Lab: SQL Server-9

Part 3: Data Definition

The main SQL data definition language statements are:

CREATE DATABASE		DROP DATABASE
CREATE DOMAIN	ALTER DOMAIN	DROP DOMAIN
CREATE TABLE	ALTER TABLE	DROP TABLE
CREATE VIEW		DROP VIEW

These statements are used to create, change, and destroy the structures that make up the conceptual schema.

SQL Data types: Character data

- CHARACTER can be abbreviated to CHAR
- CHARACTER VARYING to VARCHAR.
- CHAR(n) – fixed-length character data, *n* characters long
- VARCHAR(n) – variable length character data, with maximum length *n.*
- For example,
  - branchNo CHAR(4)
  - address VARCHAR(30)

SQL Data types: Numeric Data

- NUMERIC(p, q) – Fixed point number, with user-specified precision of *p* digits, with *q* digits to the right of decimal point.
- DEC (p, q) DECIMAL.
- INT Integer.
- SMALLINT Small integer .

Example:
salary DECIMAL(7,2) example: [12345,67]
rooms SMALLINT

SQL CREATE DATABASE Statement

1- The CREATE DATABASE statement is used to create a new SQL database.

Syntax

CREATE DATABASE *databasename*;

Example

CREATE DATABASE testDB;

2- The DROP DATABASE statement is used to drop an existing SQL database.

Syntax

```
DROP DATABASE databasename;
```

The SQL CREATE TABLE Statement

1- The CREATE TABLE statement is used to create a new table in a database.

Syntax

```
CREATE TABLE table_name (
   column1 datatype,
   column2 datatype,
   column3 datatype,
  ....
);
```

The following example creates a table called "Persons" that contains five columns: PersonID, LastName, FirstName, Address, and City:

Example 8.1 CREATE TABLE

```
CREATE TABLE Persons (
   PersonID int,
   LastName varchar(255),
   FirstName varchar(255),
   Address varchar(255),
   City varchar(255)
);
```

The PersonID column is of type int and will hold an integer.
The LastName, FirstName, Address, and City columns are of type varchar and will hold characters, and the maximum length for these fields is 255 characters.
The empty "Persons" table will now look like this:

PersonID	LastName	FirstName	Address	City

Tip: The empty "Persons" table can now be filled with data with the SQL INSERT INTO statement.

The SQL DROP TABLE Statement

1- The DROP TABLE statement is used to drop an existing table in a database.

Syntax

DROP TABLE *table_name*;

Example 8.2 CREATE TABLE
Create the Students *table using the following data.*

Student Number	Student LastName	Student FirstName	Email	PhoneNumber
190	Smith	John	jsmith@usna.edu	410-431-3456
673	Doe	Jane	jdoe@usna.edu	
312	Red	Bob	bred@usna.edu	443-451-7865

CREATE TABLE **Students**
(**StudentNumber** INTEGER,
StudentLastName CHAR (18) NOT NULL,
StudentFirstName CHAR (18) NOT NULL,
Email CHAR (50),
PhoneNumber CHAR(18));

Constraints can be defined within the CREATE TABLE statement, or they can be added to the table after it is created using the ALTER table statement. Five types of constraints:

- NOT NULL - Indicates that a column cannot store NULL value
- UNIQUE - Ensures that each row for a column must have a unique value
- PRIMARY KEY - Ensures that a column (or combination of two or more columns) have an unique identity which helps to find a particular record in a table more easily and quickly
- FOREIGN KEY - Ensure the referential integrity of the data in one table to match values in another table
- CHECK - Ensures that the value in a column meets a specific condition
- DEFAULT - Specifies a default value when specified none for this column

Adding Constraints to Students table

1) Adding constraints (PRIMARY KEY, UNIQUE) to Students table;

CREATE TABLE **Students**
(**StudentNumber** INTEGER,
StudentLastName CHAR (18) NOT NULL,

StudentFirstName CHAR (18) NOT NULL,
Email CHAR (50),
PhoneNumber CHAR(18),
PRIMARY KEY (**StudentNumber**),
UNIQUE (**Email**) *);*

2) Adding also to Students table the following constraints:

- Default value for PhoneNumber: 410-123-4567
- Email like "*@usna.edu"

CREATE TABLE **Students**
(**StudentNumber** INTEGER,
StudentLastName CHAR (18) NOT NULL,
StudentFirstName CHAR (18) NOT NULL,
Email CHAR (50),
PhoneNumber CHAR (18) DEFAULT "410-123-4567",
PRIMARY KEY (**StudentNumber**),
UNIQUE (**Email**),
CHECK (**Email** LIKE "*@usna.edu")
);

3) Adding FOREIGN KEY Constraints

The MajorDepartmentName is a foreign key referencing the Department table.

DEPARTMENTS

DepartmentName	Phone	Building	Room
Mathematics	410-293-4573	Michelson Hall	308
History	410-293-2255	Sampson Hall	120
Computer Science	410-293-6800	Michelson Hall	340

STUDENTS

Student Number	Student LastName	Student FirstName	Email	PhoneNumber	MajorDepartmentName
190	Smith	John	jsmith@usna.edu	410-431-3456	
673	Doe	Jane	jdoe@usna.edu		Computer Science
312	Doe	Bob	bred@usna.edu	443-451-7865	Mathematics

CREATE TABLE **Departments**
(**DepartmentName** CHAR(18),
Phone CHAR(18) NOT NULL,
Building CHAR(18),
Room INTEGER,
PRIMARY KEY (**DepartmentName**)
);

CREATE TABLE **Students**
(**StudentNumber** INTEGER,
StudentLastName CHAR (18) NOT NULL,

```
StudentFirstName CHAR (18) NOT NULL,
Email CHAR (50),
PhoneNumber CHAR (18) DEFAULT "410-123-4567",
MajorDepartmentName char(18),
PRIMARY KEY (StudentNumber),
UNIQUE (Email),
CHECK (Email LIKE "*@usna.edu"),
FOREIGN KEY (MajorDepartmentName)
    REFERENCES Departments (DepartmentName)
    ON DELETE SET NULL
    ON UPDATE CASCADE
);
```

SQL support all 4 options on deletes and updates.

- Default is NO ACTION (delete/update is rejected)
- CASCADE (also delete all rows that refer to deleted row)
- SET NULL / SET DEFAULT (sets foreign key value of referencing row)

Example 8.3 CREATE TABLE

Create the Employee *table using the available features of the CREATE TABLE statement.*

```
CREATE TABLE Employee
( RegNo CHAR(6),
FirstName CHAR(20) NOT NULL,
Surname CHAR (20) NOT NULL,
Dept CHAR (15),
Salary NUMERIC(9) DEFAULT 0,
City CHAR (15),
PRIMARY KEY(RegNo),
FOREIGN KEY(Dept)
    REFERENCES Department(DeptName)
    ON DELETE SET NULL
    ON UPDATE CASCADE,
UNIQUE(FirstName,Surname) );
```

The primary key is the registration number, RegNo. SQL automatically enforces uniqueness on this column. The department, Dept, is a foreign key referencing the Department table. A deletion rule has been specified such that, if a record is deleted from the Department table, the corresponding values of the Dept column in the Employee table are set to NULL. Additionally, an update rule has been specified such that, if a registration number is updated in the Department table, the corresponding values in the Dept column in the Employee table are updated accordingly. SQL assumes that the foreign keys match the primary keys of the respective parent tables.